

CAVD

→ Create ✓

→ Read

→ DISTINCT —

→ PRINT constant value —

→ Operations on Columns —

→ Inserting data from Another Table —

→ WHERE Clause ✓

→ AND, OR, NOT → && || ! ✓

→ IN operator = —

→ BETWEEN Operator —

→ LIKE —

→ IS NULL → Good Assignment Question

→ ORDER BY

→ LIMIT } → him

→ UPDATE

→ DELETE

→ Delete vs Truncate vs Drop

→ Truncate

→ Drop

Rules

→ No irrelevant questions X

→ Bonnie's

WHERE CLAUSE

Select . . . from <table>

```
if ( . . . ) {  
  |  
  }  
}
```

Where {conditions}

→ filtering logic

e.g. select all the films from film table which have rating PG-13.

Select * from film

Where (rating = 'PG-13');

→ After the from clause

Pseudo code:-

answer = [] // A list of all rows with PG-13 rating

for each row in film;

if row.matches (conditions in where clause) → WHERE
| answer.append (row);
| }
}

WHERE
clause is
analogous to
if condition

filtered_ans = [] // filtered on the basis of cols. (rating, release-year)

for each row in answer:

filtered_ans.append (row ['rating'], row ['release-year']);
| }
}

filtered-
answer =
PG-13 2006
PG-13 2006
PG-13 2007
PG-13 2008



unique-
answer =

PG-13 2006
PG-13 2007
PG-13 2008

unique-answer = set (filtered-answer);

AND, OR, NOT → logical operators in programming.

e.g. Get all films with rating PG-13 and release in 2006.

select * from films where rating = 'PG-13'

AND release-year = 2006

```
if (rating == 'PG-13' && release-year == 2006) {  
    | // pick this row  
}
```

e.g. Get all films with rating PG-13 or release in 2006.

select * from films where rating = 'PG-13'

OR release-year = 2006

NOT operator to negate a condition (!)

NOT

use this.

select * from film where NOT rating = 'PG-13'

①

select * from film where (rating = 'PG-13' OR release-year = 2006)

AND (rental-rate = 0.99)

②

select * from film where (rating = 'PG-13' OR (release-year = 2006

AND rental-rate = 0.99))

if (a == b || b == c) && d == e) — ①

if (a == b || (b == c && d == e)) — ②

comparisons $\Rightarrow$ $=$, $>$, $<$, $\geq$, $\leq$, $\neq$, $<>$

select * from film where rating <> 'PG-13';

IN Operator

e.g. I want all movies with rating PG-13, R, and G.

{ select * from film where
rating = 'PG-13' OR rating = 'R' or rating = 'G' }

select * from film where rating IN ('PG-13', 'G', 'R');

BETWEEN Operator

Get all films which have release year ≥ 2005 and ≤ 2010

select * from films where year ≥ 2005 and year ≤ 2010 ;

→ select * from films where year BETWEEN 2005 AND 2010
↓
inclusive of the values.

LIKE operator

Scaler DB

DSM2 → Da,
Academy → swe1

Batches

name (String Data Type)

Academy Intermediate

DSM2 Advance

Academy Advan.

Intermediate Dec 2022 Academy.

Academy
Academy } →

select * from Batches WHERE name LIKE %Academy%

↓
Allows charac on left side.

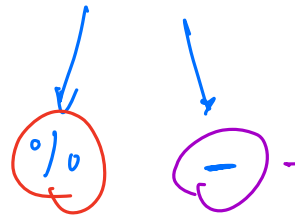
Int Academy //

Adv Academy //

Dec Jan Academy //

Jan Academy Intermediate //

How LIKE operator works?


It matches exactly 1 character
Matches ≥ 0 any set of character
% LOVE
PARIS LOVE

① cat% → cat, caterpillar, category, cat
Not match with wildcat, dog.

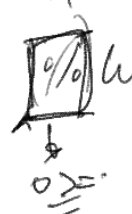
② %cat → wildcat, domestic cat, cat
Not match with cattle

③ %cat% → cat,

④ _at → cat, bat, hat, fat
matches 1 char
X wildcat, chest X

⑤ c_t → cat, cot, cut
X chat, cont X, cap X, hot X

⑥ c%ot → cat, cot, cut, chart, connect, ct
Any no. of char in between
X wildcat, caterpillar X.

He loves Su

0 >= 0

IS NULL →

NULL → empties

select * from film where rating = NULL; X

↓
can't use this

Why this query will not return any row?

→ NULL is not equal to NULL
NULL is not equal to anything

select NULL = NULL; → NULL

3 = NULL; → NULL

3 <> NULL; → NULL

* We compare anything with NULL using IS NULL operator.

select * from film where title IS NULL;

select * from film where title IS NOT NULL;

e.g. Find customers with id other than 2;

NULL X
id != 2.

select * from customers where id != 2;

→ You are missing the customer rows with id = NULL.

→ This will not return the customer with id NULL.

id
2 → id != 2 X
3 → id != 2 ✓
4 → id != 2 ✓
5 → id != 2 ✓
2 → id != 2 X
9 → id != 2 ✓
NULL → NULL != 2 NULL

True AND NULL

NOT NULL

Correct query:-

select * from customers where id IS NULL OR $id \neq 2$.

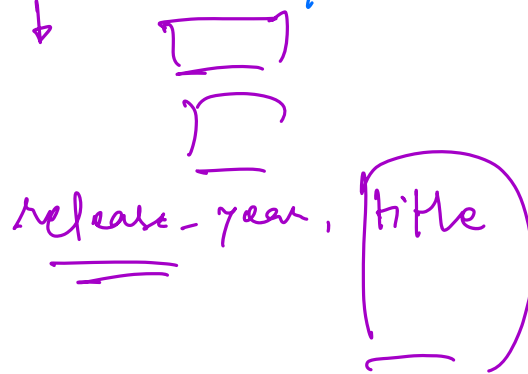
id IS NULL OR $id \neq 2$ X
↓
NULL

ORDER BY CLAUSE

select * from film order by title; (by default ascending)

select * from film order by title desc; (descending order)

select * from film order by title, release-year;



Pseudo Code:-

answer = [];

for each row in film:

if row.matches(where conditions):
answer.append(row);

WHERE

length, title

answer.sort(column-names in order by clause) ORDER BY

filtered-ans = []

for each row in answer:

filtered-ans.append(ans[rating], row[release])

SELECT

1st September → Friday 9 PM } → Please be present
↳ Class

Tues, Thur, Fri, Sat →

F_Name	L_Name
Alok	Singh
Vijay	Kumar
Alok	Kumar
Vijay	Singh.

Alok	Kumar
Alok	Singh.
Vijay	Ku
Vijay	Singh.

How to alter a table to create a new column

"retail." →



like xps%

retail_xps.

retail

1 = ~~id~~hps%
0

PK

1

2

FName LName

Alok Singh

Alok Singh.

}

If every thing similar
sort by PK